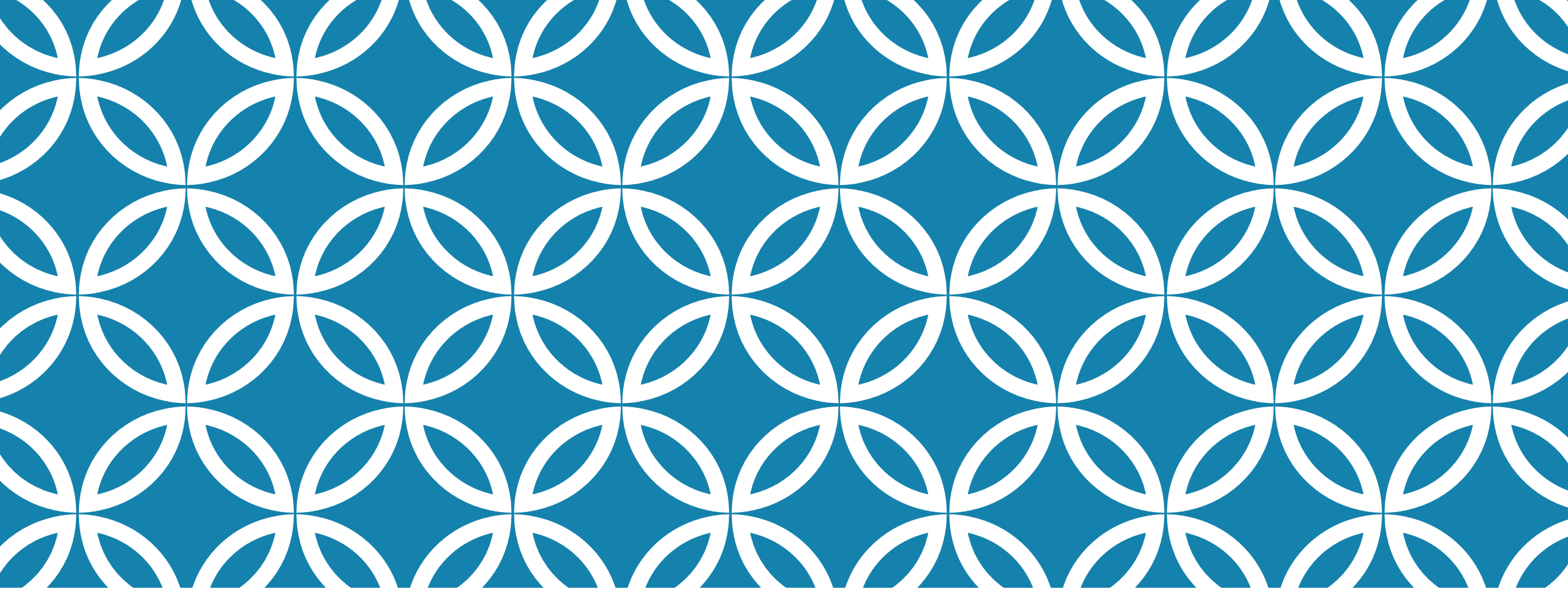




SEC8_DATA SCIENCE

Eng: mariam khashapa

CHEST X-RAY CLASSIFICATION (NORMAL VS PNEUMONIA)

Data folder

dataset/

train/

NORMAL/

PNEUMONIA/

val/

NORMAL/

PNEUMONIA/

<https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>

Importing libraries

```
import tensorflow as tf
from tensorflow.keras import layers, models
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import os
import cv2
```

Loading data

```
IMG_SIZE = 128
def load_data(path):
    images = []
    labels = []
    for label in ["NORMAL", "PNEUMONIA"]:
        folder = os.path.join(path, label)
        for img_name in os.listdir(folder):
            img_path = os.path.join(folder, img_name)
            img = cv2.imread(img_path)
            if img is None:
                continue
            img = cv2.resize(img, (IMG_SIZE, IMG_SIZE))
            img = img / 255.0

            images.append(img)
            labels.append(0 if label == "NORMAL" else 1)
    return np.array(images), np.array(labels)
X_train, y_train = load_data("dataset/train")
X_val, y_val = load_data("dataset/val")

print("Train shape:", X_train.shape)
print("Validation shape:", X_val.shape)
```

```
df = pd.DataFrame({  
    "label": y_train  
})
```

```
print(df["label"].value_counts())
```

```
plt.figure(figsize=(10,5))
```

```
for i in range(6):  
    plt.subplot(2,3,i+1)  
    plt.imshow(X_train[i])  
    plt.title("Normal" if y_train[i]==0 else "Pneumonia")  
    plt.axis("off")
```

```
plt.show()
```

Building the model

```
model = models.Sequential([
    layers.Conv2D(32, (3,3), activation='relu',
input_shape=(128,128,3)),
    layers.BatchNormalization(),
    layers.MaxPooling2D(),
    layers.Conv2D(64, (3,3), activation='relu'),
    layers.BatchNormalization(),
    layers.MaxPooling2D(),
    layers.Conv2D(128, (3,3), activation='relu'),
    layers.BatchNormalization(),
    layers.MaxPooling2D(),
    layers.Conv2D(256, (3,3), activation='relu'),
    layers.MaxPooling2D(),
    layers.Flatten(),
    layers.Dense(256, activation='relu'),
    layers.Dropout(0.5),
    layers.Dense(1, activation='sigmoid')
])
```

```
model.compile(  
    optimizer='adam',  
    loss='binary_crossentropy',  
    metrics=['accuracy']  
)
```

```
model.summary()
```

```
history = model.fit(  
    X_train, y_train,  
    validation_data=(X_val, y_val),  
    epochs=15,  
    batch_size=32  
)
```

```
# Accuracy
plt.plot(history.history['accuracy'], label='train')
plt.plot(history.history['val_accuracy'], label='val')
plt.title("Accuracy")
plt.legend()
plt.show()
```

```
# Loss
plt.plot(history.history['loss'], label='train')
plt.plot(history.history['val_loss'], label='val')
plt.title("Loss")
plt.legend()
plt.show()
```

```
loss, acc = model.evaluate(X_val, y_val)
print("Validation Accuracy:", acc)
```



```
img = X_val[0]
prediction = model.predict(np.expand_dims(img, axis=0))

print("Prediction:", "Pneumonia" if prediction[0][0] > 0.5 else
      "Normal")

plt.imshow(img)
plt.title("Prediction Result")
plt.axis("off")
plt.show()
```

Task

Build a deep learning model using TensorFlow to classify brain MRI images into different tumor categories.

Link for data

<https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>